

Calibration results

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Normalized Residuals

Reprojection error (cam0): mean 0.06149309961064179, median 0.05731708337180669, std: 0.03293517804472805
Gyroscope error (imu0): mean 9.404923504181665e-10, median 3.0856098674273706e-10, std: 2.4634479550481947e-09
Accelerometer error (imu0): mean 1.0437891490509096e-09, median 7.39844892852023e-11, std: 1.1460645248190959e-08

Residuals

Reprojection error (cam0) [px]: mean 0.06149309961064179, median 0.05731708337180669, std: 0.03293517804472805
Gyroscope error (imu0) [rad/s]: mean 5.395063684924089e-13, median 1.7700369103692485e-13, std: 1.4131384052269806e-12
Accelerometer error (imu0) [m/s^2]: mean 7.395392194556749e-12, median 5.241904603775163e-13, std: 8.120027544846744e-11

Transformation (cam0):

T_ci: (imu0 to cam0):

[[-0.01975654 -0.99963268 -0.01855226 -0.00000008]
[0.041435 0.0177213 -0.99898403 0.00000039]
[0.99894585 -0.02050518 0.04106967 -0.00000032]
[0. 0. 0. 1.]]

T_ic: (cam0 to imu0):

[[-0.01975654 0.041435 0.99894585 0.00000003]
[-0.99963268 0.0177213 -0.02050518 -0.00000009]
[-0.01855226 -0.99898403 0.04106967 0.00000004]
[0. 0. 0. 1.]]

timeshift cam0 to imu0: [s] (t_imu = t_cam + shift)
-0.00019224072595731714

Gravity vector in target coords: [m/s^2]

[-0.08218228 1.7355133 -9.65140728]

Calibration configuration

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cam0

Camera model: pinhole

Focal length: [387.8745225532805, 387.71007959205747]

Principal point: [327.0625973660385, 240.49153036139765]

Distortion model: radtan

Distortion coefficients: [-0.053472001544476296, 0.05238894707284819, 0.0004985375447714819,
0.00046666320057901675]

Type: checkerboard

Rows

Count: 8

Distance: 0.02 [m]

Cols

Count: 11

Distance: 0.02 [m]

IMU configuration

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IMU0:

Model: calibrated

Update rate: 50.0

Accelerometer:

Noise density: 0.0010019901002151698

Noise density (discrete): 0.007085139945439349

Random walk: 1.9893785362909353e-05

Gyroscope:

Noise density: 8.112529814510266e-05

Noise density (discrete): 0.0005736424844418254

Random walk: 6.28721982800146e-07

T_ib (imu0 to imu0)

[[1. 0. 0. 0.]

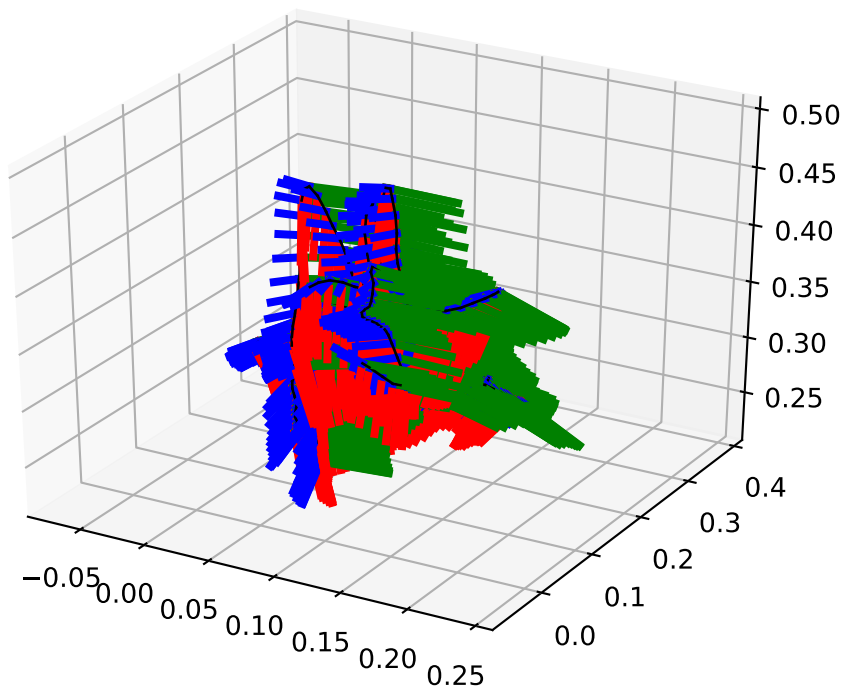
[0. 1. 0. 0.]

[0. 0. 1. 0.]

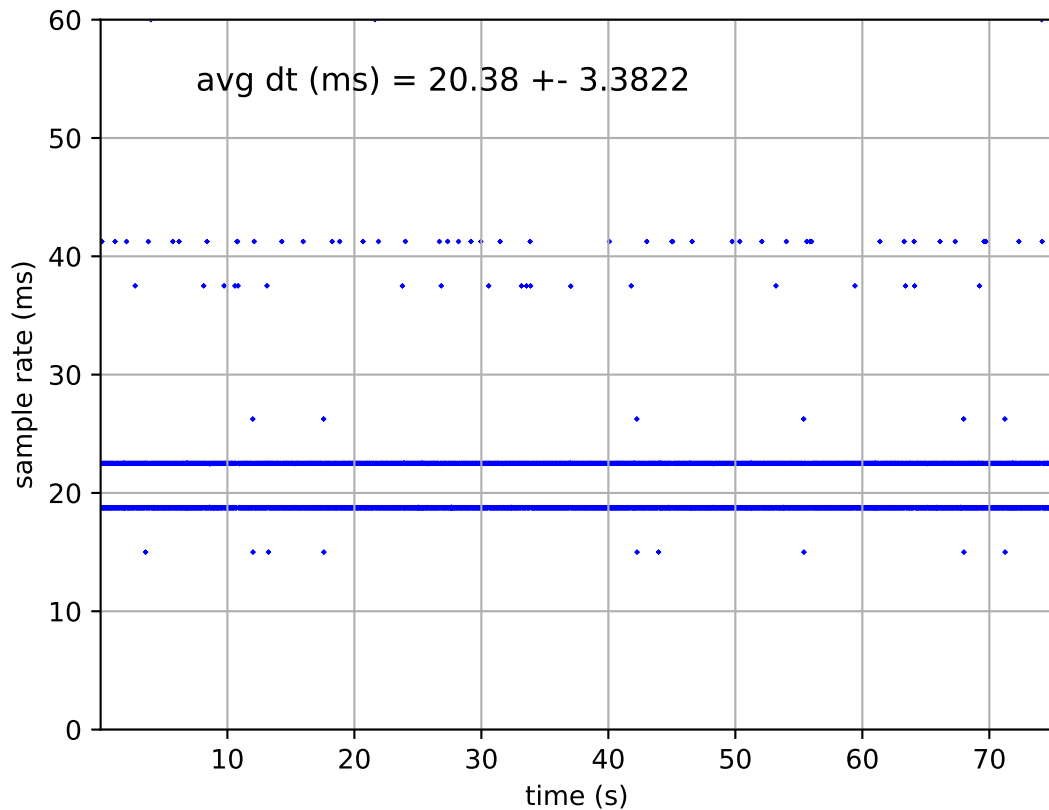
[0. 0. 0. 1.]]

time offset with respect to IMU0: 0.0 [s]

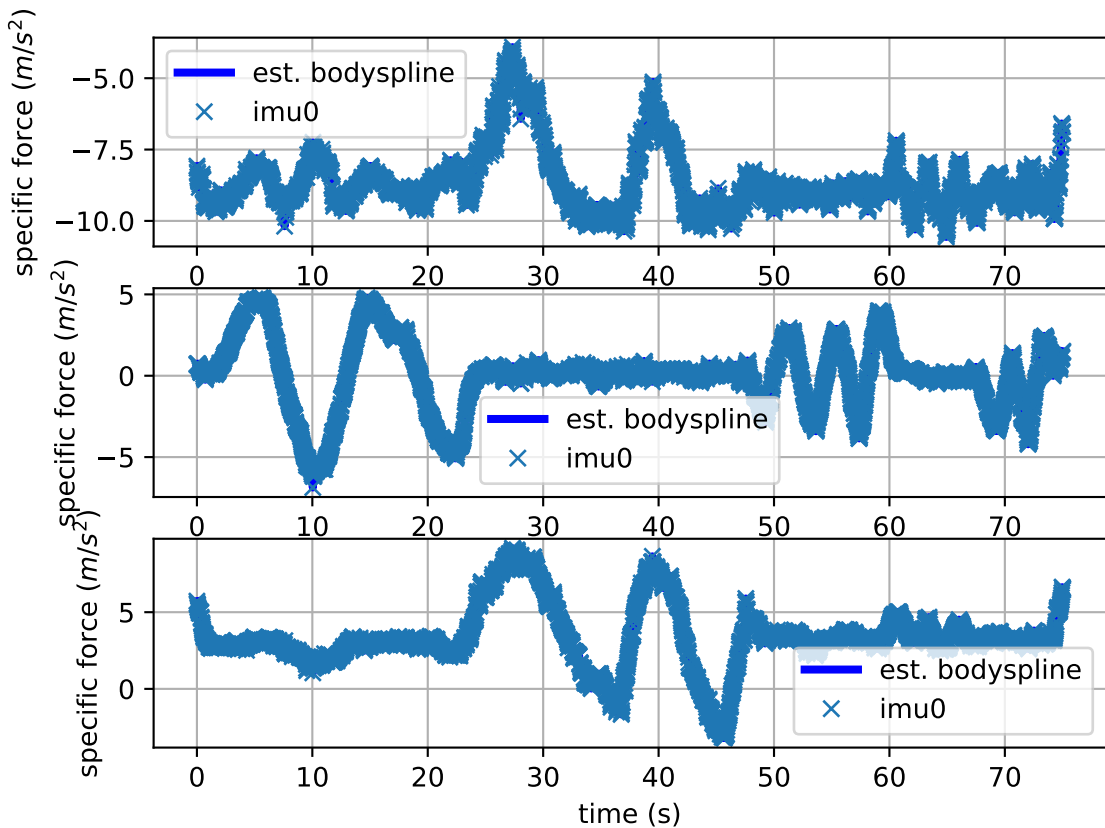
imu0: estimated poses



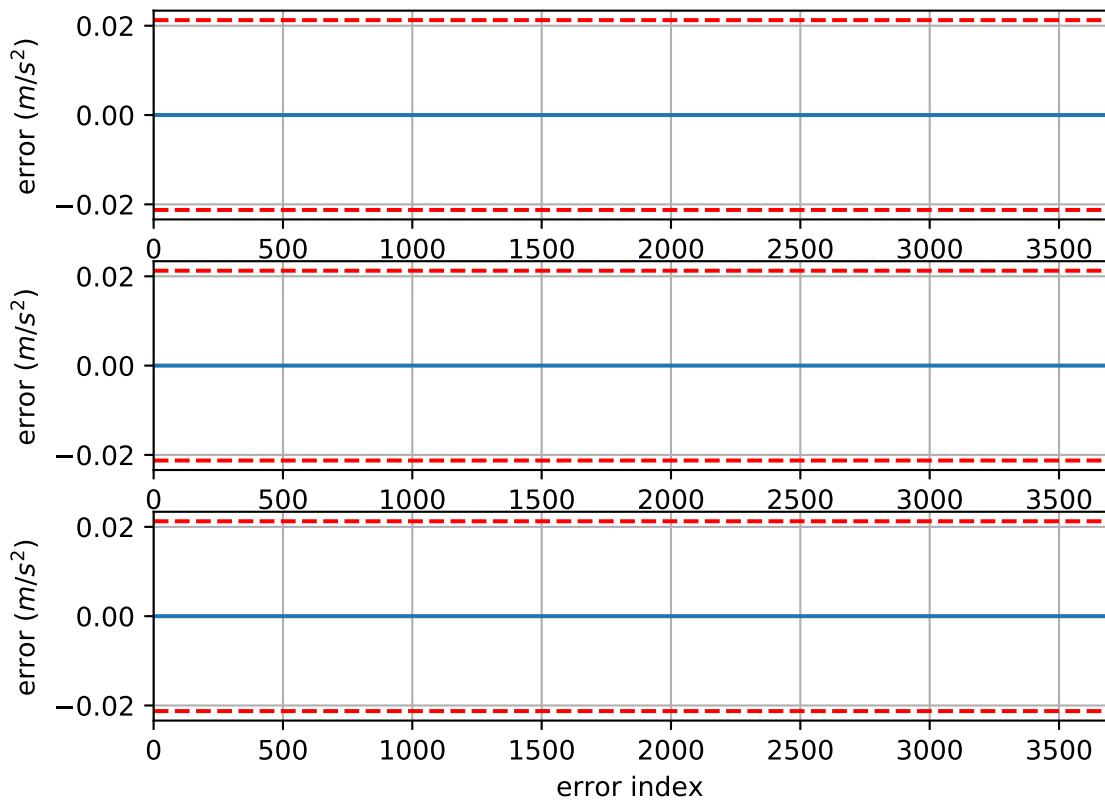
imu0: sample inertial rate



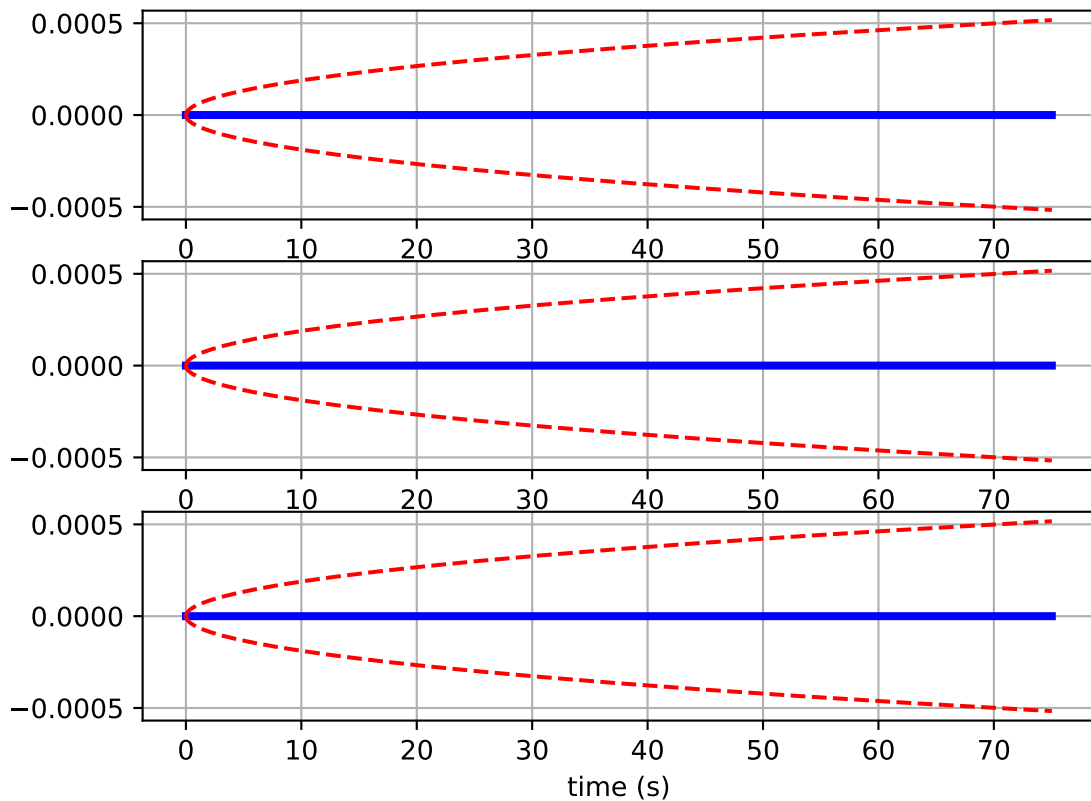
Comparison of predicted and measured specific force (imu0 frame)



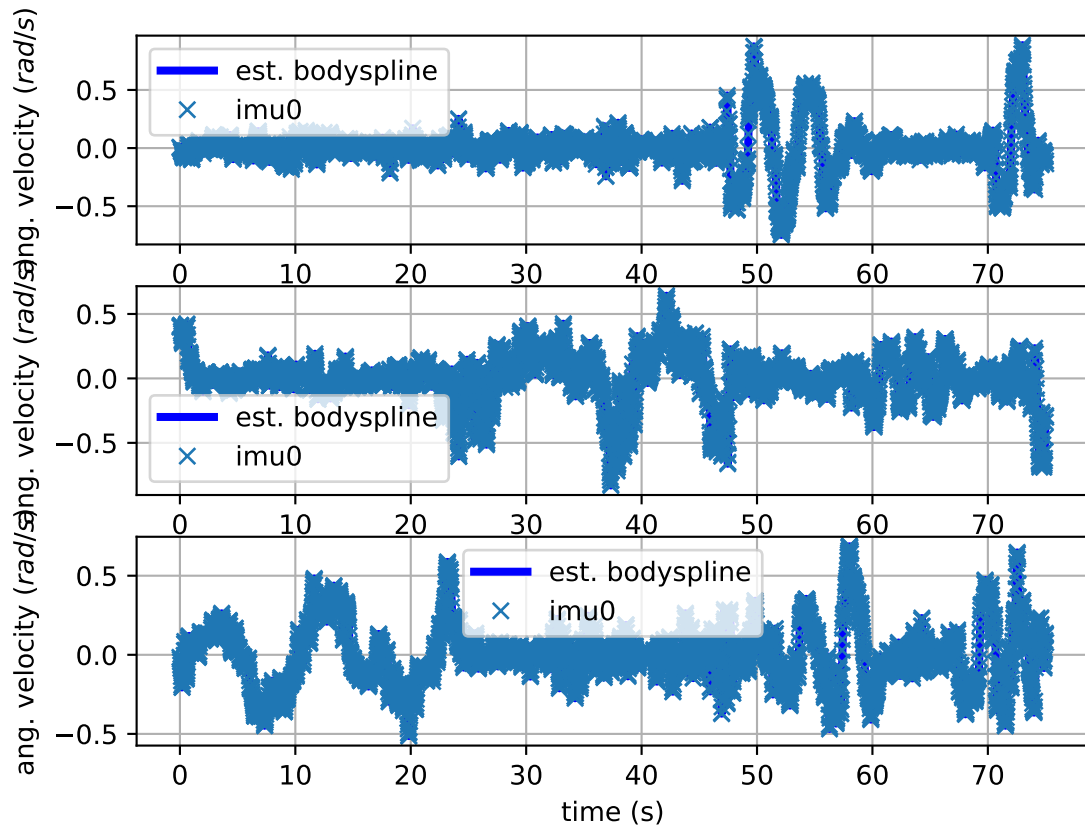
imu0: acceleration error



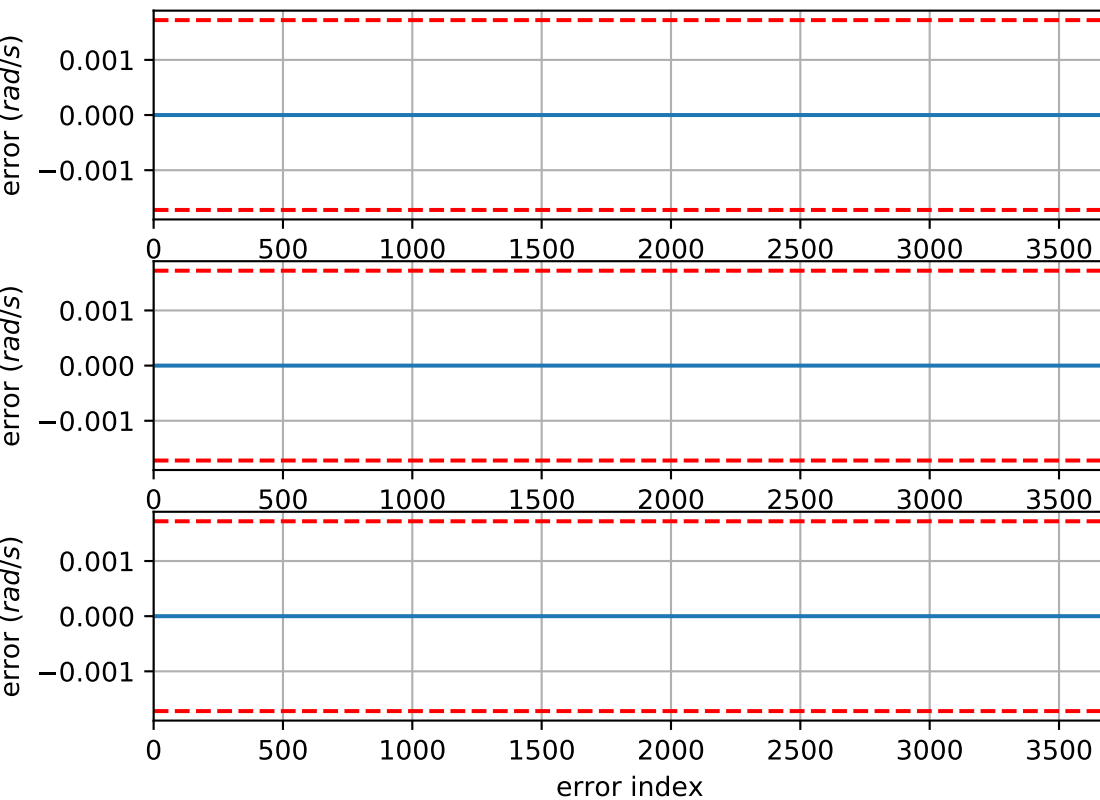
imu0: estimated accelerometer bias (imu frame)



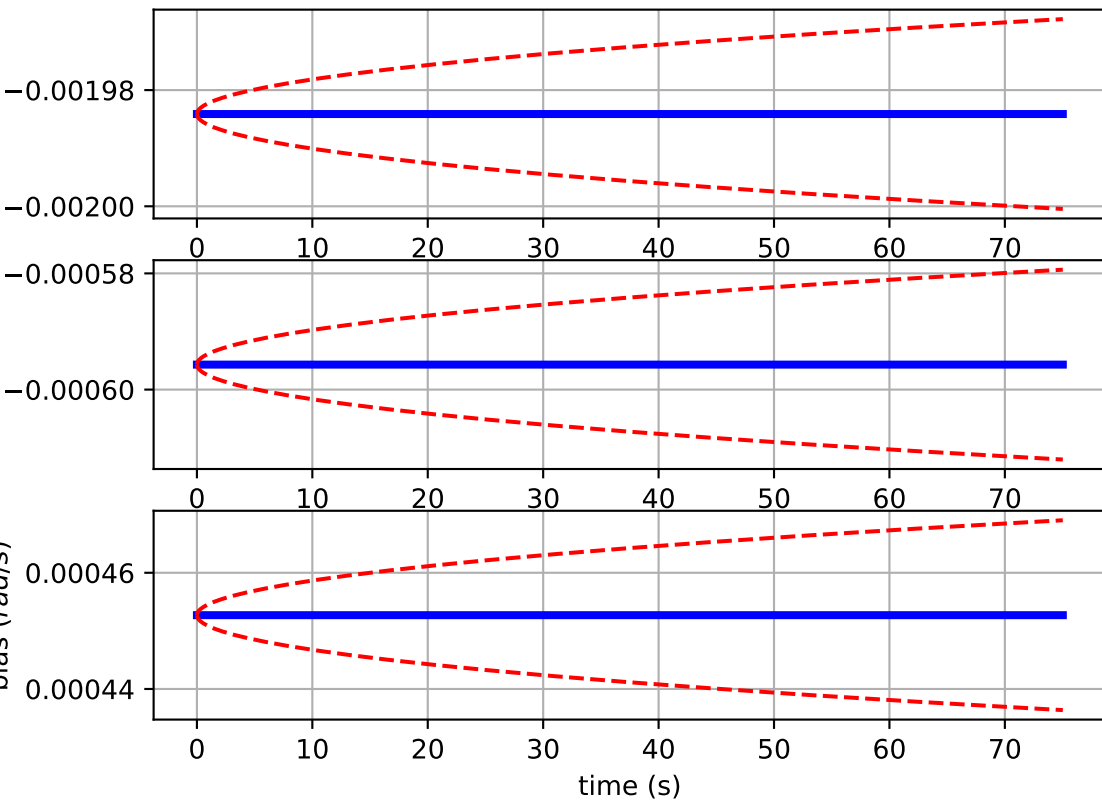
Comparison of predicted and measured angular velocities (body frame)



imu0: angular velocities error



imu0: estimated gyro bias (imu frame)



cam0: reprojection errors

